

Beginning Algebra Test II

Name
Date
Course

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show work when necessary.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.

Simplify.

① $\frac{x^2+4x-32}{x^2-16}$ ② $\frac{16x^2-25}{4x-5}$ ③ $\frac{8x-4x^2}{24x^4-64x^3+32x^2}$

Perform the indicated operations.

④ $\frac{3x-y}{x^2-1} \cdot \frac{x+1}{6x^2+25xy-9y^2}$ ⑤ $\frac{x^2+x-12}{x-2} \div \frac{9-x^2}{3x^2-6x}$

⑥ $\frac{5}{x-11} + \frac{2x}{x^2-10x-11}$ ⑦ $\frac{6}{5x} - \frac{x-8}{x(x-1)}$

Find the square roots. You will have to approximate one of the roots with a calculator (Round to the nearest thousandth).

⑧ a) $\sqrt{\frac{4}{9}}$ b) $\sqrt{49}$ c) $\sqrt{10}$ d) $\sqrt{1}$

⑨ Factor out a perfect square: a) $\sqrt{18}$ b) $\sqrt{112}$

Add or subtract: c) $3\sqrt{5}-8\sqrt{5}+2\sqrt{5}$

d) $-4\sqrt{2}-\sqrt{2}+(-3\sqrt{2})$

Multiply. Never leave perfect square factors under the root signs.

⑩ a) $\sqrt{3} \cdot \sqrt{3}$ b) $-\sqrt{5} \cdot 2\sqrt{3}$ c) $5\sqrt{6}(-2\sqrt{3})$
d) $8\sqrt{6}(4\sqrt{2}+2\sqrt{5}-2\sqrt{6})$

⑪ Divide or multiply: a) $-2(-4\sqrt{6})$ b) $\frac{\sqrt{24}}{\sqrt{3}}$

Rationalize the denominators: c) $\frac{3}{\sqrt{7}}$ d) $\frac{6}{\sqrt{27}}$

Simplify.

⑫ a) $\frac{\frac{3}{8}}{\frac{1}{4}}$ b) $\frac{\frac{7}{2}}{\frac{2}{5}}$ c) $\frac{-\frac{4x}{5}}{3}$ d) $\frac{\frac{2}{3}+\frac{1}{6}}{\frac{1}{4}-\frac{1}{2}}$

Divide.

⑬ $\frac{15x^4-13x^2-20}{30x^5-6x^6} \div \frac{-50x^6-40x^4}{-50x+2x^3}$

Simplify (i.e. add and/or subtract).

⑭ a) $-3\sqrt{20}+2\sqrt{10}-\sqrt{5}-8\sqrt{90}$

Factor out a perfect square(s) (leave no perfect square factor in the root).

b) $\sqrt{128}$

Rationalize the denominators.

(15) a) $\sqrt{\frac{1}{9}}$ b) $\sqrt{\frac{2}{13}}$ c) $\sqrt{\frac{100}{7}}$ d) $\sqrt{\frac{11}{25}}$

Simplify.

Rationalize the denominator.

(16) a) $\frac{-2}{x-3} \div \frac{4}{5x}$

b) $\sqrt{\frac{10}{99}}$

Simplify.

(17) $\frac{\frac{5x}{3y^2}}{\frac{z}{6yx^2}}$

Add and/or subtract.

(18) $-\frac{2}{3b} + \frac{1}{b-4} - \frac{7}{b}$

Rationalize the denominators.

(19) a) $\sqrt{\frac{28}{11}}$ b) $\sqrt{\frac{45}{32}}$

Add.

(20) $\frac{8}{x^2+4x+4} + \frac{1}{x^2+3x+2}$

Add and/or Subtract.

(21) $\frac{4}{1-x} - \frac{x+3}{x+1} + \frac{1}{x^2-1}$

Add and/or Subtract.

(22) $\frac{5x+6}{2x^2-13x-24} - \frac{x-7}{2x+3} + \frac{3x}{x-8}$

Simplify.

(23) $\frac{\frac{6}{x^2-4x-5}}{\frac{-2}{x^2-5x}}$

(24) $\frac{\frac{2}{x-3} + \frac{1}{x-7}}{-2 + \frac{x+2}{x^2-10x+21}}$

Add and/or subtract.

(25) $\frac{-8x-3}{9x^2-16} - \frac{6}{4-3x} + 2$

Beginning Algebra Test II Answers

① $\frac{x+8}{x+4}$

② $4x+5$

③ $\frac{-1}{2x(3x-2)}$ or $\frac{1}{2x(2-3x)}$ or $-\frac{1}{2x(3x-2)}$

④ $\frac{1}{(2x+9y)(x-1)}$

⑤ $\frac{-3x(x+4)}{x+3}$ or $\frac{3x(x+4)}{-(x+3)}$ or $-\frac{3x(x+4)}{x+3}$

⑥ $\frac{7x+5}{(x-11)(x+1)}$

⑦ $\frac{x+34}{5x(x-1)}$

⑧ a) $\frac{2}{3}$ b) 7 c) 3.162 d) 1

⑨ a) $3\sqrt{2}$ b) $4\sqrt{7}$ c) $-3\sqrt{5}$ d) $-8\sqrt{2}$

⑩ a) 3 b) $-2\sqrt{15}$ c) $-30\sqrt{2}$ d) $64\sqrt{3} + 16\sqrt{30} - 96$

⑪ a) $8\sqrt{6}$ b) $2\sqrt{2}$ c) $\frac{3\sqrt{7}}{7}$ d) $\frac{6\sqrt{3}}{9} = \frac{2\sqrt{3}}{3}$

⑫ a) $\frac{3}{2}$ b) $\frac{35}{2}$ or $17\frac{1}{2}$ c) $-\frac{4x}{15}$ d) $-\frac{10}{3}$ or $-3\frac{1}{3}$

⑬ $\frac{(3x^2-5)(x+5)}{30x^8}$

⑭ a) $-22\sqrt{10} - 7\sqrt{5}$

b) $8\sqrt{2}$

⑮ a) $\frac{1}{3}$ b) $\frac{\sqrt{26}}{13}$ c) $\frac{10\sqrt{7}}{7}$ d) $\frac{\sqrt{11}}{5}$

⑯ a) $\frac{-5x}{2x-6}$ or $\frac{-5x}{2(x-3)}$ b) $\frac{\sqrt{110}}{33}$

⑰ $\frac{10x^3}{yz}$

⑱ $\frac{-20b+92}{3b(b-4)}$

⑲ a) $\frac{2\sqrt{77}}{11}$ b) $\frac{3\sqrt{10}}{8}$

⑳ $\frac{9x+10}{(x+1)(x+2)^2}$ or $\frac{9x+10}{(x+1)(x+2)(x+2)}$

㉑ $\frac{-x^2-6x}{(x-1)(x+1)}$ or $\frac{-x^2-6x}{x^2-1}$

㉒ $\frac{5x^2+29x-50}{(2x+3)(x-8)}$

23 $\boxed{-\frac{3x}{x+1}}$ or $\boxed{\frac{-3x}{x+1}}$

24 $\boxed{\frac{3x-17}{-2x^2+21x-40}}$

25 $\boxed{\frac{18x^2+10x-11}{9x^2-16}}$ or $\boxed{\frac{18x^2+10x-11}{(3x-4)(3x+4)}}$